

EDUCATION

PURDUE UNIVERSITY

MS in Computer Science

Machine Learning,
NLP and Optimization theory

Grad. May 2016 | West Lafayette, IN

Cum. GPA: 3.74

R.V COLLEGE OF ENGINEERING

BEng in Computer Engineering

Grad. May 2013 | Bangalore, India

Conc. in Computer Engineering

Minor in Electrical Engineering

College of Engineering

Cum. GPA: 9.15 / 10

LINKS

Github:// [ninja91](#)

LinkedIn:// [nitinajain](#)

Scholar:// [Google Scholar](#)

Twitter:// [@anitinjain](#)

COURSEWORK

GRADUATE

Numerical Linear Algebra

Statistical Machine Learning

Machine Learning Methods For NLP

Compilers & Programming Systems

Graph Theory

Computational Methods In Optimization

Bayesian Applied Decision Theory

UNDERGRADUATE

Digital Signal Processing

Digital Communication

Linear Algebra (*Research Asst.*)

Information Theory

Computer Architecture

Data Structures and Algorithms

SKILLS

PROGRAMMING

Over 5000 lines:

Python • C++ • Java

Shell • $\text{\LaTeX}$

Over 1000 lines:

Matlab • JavaScript • Android • R •

Assembly

Familiar:

CSS • MySQL

EXPERIENCE

META | Machine Learning Engineer

Sep 2022 - Present | Seattle, WA

- Led the end-to-end ML lifecycle from EMG research to production model inference, culminating in the launch of the **Meta Neural Band** device; engineered training recipes, standardized evaluation suites, and hardware-emulator-in-the-loop benchmarking pipelines.
- Spearheaded research and production of on-device quantization, sparsity, compression, and power optimization for EMG and **handwriting models**; developed sub-8-bit/sparse models targeting strict power/latency budgets.
- Contributed quantization and inference optimizations to open-source **PyTorch** & **ExecuTorch** in mixed precision regimes (fp16/int8/int4) and for Attention, RNN modules.
- Collaborated with ARM to extend ExecuTorch, enabling quantization, lowering, and delegation of complex models to accelerators at peak performance.

AMAZON.COM | Senior Software Engineer, Alexa

June 2016 – Sep 2022 | Seattle, WA

- Led the development of a proactive, ML-driven latent goal-solving service scaling to 200+ TPS, doubling system throughput through lazy caching and rule engine optimizations.
- Launched an industry-first contextual understanding solution under Alexa's NLU service for multi-turn interactions, optimizing cloud model inference latency and compute footprint.

CISCO SYSTEMS | Software Development Engineer

Aug 2013 – Aug 2014 | Bangalore, India

- Created a patentable Face detection based Auto-Focus algorithm with Non-linear regression

RESEARCH

PURDUE UNIVERSITY | NLP Research | Prof. Goldwasser

Jan 2015 – May 2015 | West Lafayette, IN

Developed a hierarchical model to extract sports events and tag entities from Twitter's streaming API, curating a dataset of 1,000+ classified tweets.

PURDUE UNIVERSITY | Optimization & Matrix computations Research

Aug 2015 – May 2016 | West Lafayette, IN

Researched kernel learning algorithms over convex spaces (Prof. Honorio) and designed memory-efficient structures to extract a 150GB DNS census bipartite graph (Prof. Gleich).

R.V COLLEGE OF ENGINEERING | Undergraduate Researcher

Jun 2012 – May 2013 | Bangalore, India

Researched fast numerical methods to compute the n^{th} root of a number.

PATENT

USER-SYSTEM DIALOG EXPANSION | U.S. Patent 11,527,237

Granted Dec 2022 | [Patent Link](#)

Recommending skill experiences to a user after a dialog session has concluded using machine learning models to identify and rank candidate intents.

PUBLICATIONS

- [1] N. Jain, K. Murthy, and Hamsapriye. Matrix methods for finding n^{th} root of m . *IJMEST*, 2014.
- [2] N. Jain, K. Murthy, and Hamsapriye. On finding n^{th} root of m leading to newton raphson's improved method. *IJETCAS*, 2014.